# CSA2001: Fundamentals in AI and ML

VIT BHOPAL

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# VIT Bhopal Campus Navigator. Project Report

## 1. Introduction

For campuses like VIT Bhopal it is really hard for students to get from one place to another on campus especially when they first start school. Even though we have maps they do not always show us the way to go.

The goal of the VIT Bhopal Campus Navigator project is to make a system that helps students find the shortest way to get between two places on campus.

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## 2. Problem Statement

The problem the VIT Bhopal Campus Navigator project is trying to solve is:

“How can we find the way to get between two places on campus using real information?”

This problem is important because the VIT Bhopal Campus Navigator project saves time for students.

The VIT Bhopal Campus Navigator project helps students move around campus easily especially when they first start school.

The VIT Bhopal Campus Navigator project can also be used to solve real-world problems.

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# 3. Approach and Methodology

Step 1: Data Storage

We use SQLite to store information about the routes on campus. Each route has the following information:

the starting point of the route

the ending point of the route

the distance between the starting and ending points

the number of people between the starting and ending points

This makes the VIT Bhopal Campus Navigator system flexible.

The VIT Bhopal Campus Navigator system can change at any time.

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Step 2: Graph Representation

We think of the campus as a graph. The places on campus are like points. The routes are like lines that connect them. The distance is like a weight that we give to each line.

Places on campus = points

Routes = lines that connect them

Distance = weight

This lets the VIT Bhopal Campus Navigator system use a kind of math.

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Step 3: Shortest Path Algorithm

The VIT Bhopal Campus Navigator system uses the algorithm to find the shortest way to get between two places on campus.

We chose this algorithm because it works well with graphs.

The Dijkstra algorithm always finds the way.

The Dijkstra algorithm is good for medium-sized projects.

The result is shown step by step. It is easy to understand because it is in a table.

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## 4. Key Design Decisions

Some important decisions I made while working on the VIT Bhopal Campus Navigator project are:

I chose SQLite of a file system because it is more organized.

I chose NetworkX because it is good for working with graphs.

I made the input simple so that it is easy for users to understand the VIT Bhopal Campus Navigator project.

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## 5. Challenges Faced

1. Understanding Graph Concepts

It took me a while to understand how to think of places on campus as a graph.

2. Database Integration

It was a little hard to combine SQL data with Python logic for the VIT Bhopal Campus Navigator project.

3. Error Handling

It was a little hard to handle the case when there is no route between two places on campus for the VIT Bhopal Campus Navigator project.

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## 6. Learning Outcomes

What I learned from the VIT Bhopal Campus Navigator project is:

how to think of real-world problems as a graph

how to use the algorithm

how to combine databases with Python for the VIT Bhopal Campus Navigator project

how to keep code clean and easy to understand

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## 7. Limitations

The VIT Bhopal Campus Navigator project does not take into account how many people are on the route.

The VIT Bhopal Campus Navigator project only has a text-based interface, not a graphical one.

The VIT Bhopal Campus Navigator project only works for a number of places on campus.

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## 8. Future Scope

What can be done next with the VIT Bhopal Campus Navigator project is:

how to take into account how many people are on the route

how to add an interface

how to combine it with real-time location tracking for the VIT Bhopal Campus Navigator project

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9. The VIT Bhopal Campus Navigator project shows that we can use AI search algorithms and databases to solve real-world problems.

Even though it is a project it can be a good starting point, for more advanced projects.

I learned a lot from the VIT Bhopal Campus Navigator project and I got to use what I had studied.

The VIT Bhopal Campus Navigator project was an experience.